

Cross-Lingual RAG Asymmetry and Multilingual Entity Resolution Discrepancies in Large Language Models

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Abstract—This paper investigates the boundaries of AI Visibility and cross-lingual entity resolution within modern Retrieval-Augmented Generation (RAG) frameworks. Through a single-subject comparative experiment utilizing Google’s Gemini in AI Mode, we demonstrate how asymmetric training and retrieval corpora across different languages result in contradictory professional identity mapping. While a high-density language-specific footprint triggers successful entity recognition alongside hallucinated source attributions, a weak linguistic footprint leads to a false declaration of nonexistence and active automated redirect toward competitors. These findings imply that AI Visibility is highly language-fragmented, posing novel brand safety risks for international digital entities.

Index Terms—Generative Engine Optimization (GEO), AI Visibility, Entity SEO, Large Language Models, Retrieval-Augmented Generation (RAG), Cross-Lingual Asymmetry.

I. INTRODUCTION

In the era of generative search, the paradigm of search engine optimization is shifting from URL indexation to machine-readable entity comprehension. While classical information retrieval focused on page relevance, modern generative answer engines, including Google AI Overviews, utilize Retrieval-Augmented Generation (RAG) to synthesize direct answers. This paper explores a critical, yet unmeasured business risk: the linguistic fragmentation of professional identity and brand safety within identical LLM architectures when queried across different language corpora.

II. METHODOLOGY

A comparative empirical experiment was conducted in August 2026 to assess identity mapping behavior. Over the preceding six weeks, a concentrated digital footprint was established in the Russian-language segment, consisting of five high-density technical articles regarding Generative Engine Optimization (GEO).

Two queries were submitted minutes apart to Google’s Gemini (AI Mode), using equivalent structural search intents separated only by language and script translation:

- **Query 1 (Russian):** Владимир Синицын geo
- **Query 2 (English):** vladimir sinitsyn seo

The outputs were evaluated side by side across multiple vectors: identity resolution accuracy, total citations returned, factual fidelity of source attribution, and competitive recommendation behavior.

III. RESULTS AND EMPIRICAL ANALYSIS

The experiment revealed directly contradictory outputs from the same model instance based strictly on the language of the query. A side-by-side metric matrix is provided in Table I.

Table I
SIDE-BY-SIDE RAG EVALUATION MATRIX

Metric Layer	Russian Corpus (RU)	English Corpus (EN)
Identity Verdict	Verified SEO Specialist	Nonexistent Expert
Total Citations	6 Citations	6 Citations
Accurate Links	4 Links Verified	0 Links Verified
Fabricated Links	2 Attributions Fabricated	0 Attributions
Competitor Shift	None	Redirected to 3 Competitors

In the Russian segment, Gemini performed successful disambiguation and generated a detailed professional overview. However, 33.3% of the citations were fabricated attributions—linking to valid external GEO materials not authored by the subject.

In the English segment, the model initiated a hard denial of prominence, resolved the proper noun to four unrelated cross-industry individuals, and actively recommended three competing figures under the “Possible Explanations for Your Search” layout block.

IV. DISCUSSION AND RAG ASYMMETRY HYPOTHESIS

The observed anomalies support the hypothesis of **cross-lingual RAG asymmetry combined with weak entity resolution boundaries**.

In a dense language corpus, when content density crosses an identification threshold, the RAG mechanism retrieves adjacent topical chunks and aggregates them under the dominant

entity vector via topical proximity. This explains the presence of citation-backed hallucinations.

Conversely, in an empty language corpus, the lack of interconnected entity signals causes the system to map the query to alternative vector coordinates (unrelated individuals) and fill the contextual semantic gap (the “SEO” token) with established market leaders.

V. CONCLUSION AND STRATEGIC GEO FRAMEWORK

AI Visibility is not a singular metric; it is language-fragmented. To maintain brand safety and professional authority in generative systems, founders and experts must deploy multi-market entity architectures. The mitigation of RAG asymmetry requires:

- 1) Concentrated density-first publishing strategies rather than multi-language dilution.
- 2) Strict implementation of machine-readable structured microdata (`Schema.org/Person`).
- 3) Deployment of root-level LLM communication architectures, specifically optimized `llms.txt` indexing assets.

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